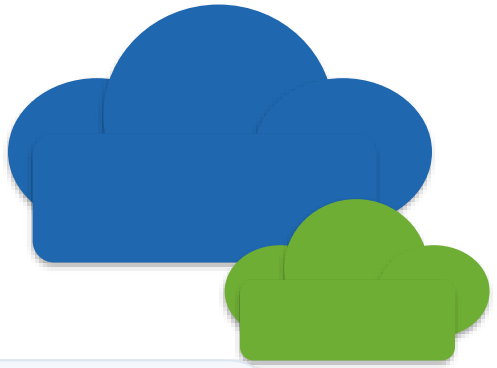
SIMATS ENGINEERING, CHENNAI

MULTI-CLOUD AND HYBRID CLOUD STRATEGIES

FOR ENTERPRISE IT

Deployment models • Business benefits • Challenges • Practical examples



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Academic Presentation • Prepared in a Microsoft Office-style layout

Introduction & Deployment Models

Two strategies help organizations balance flexibility, control, resilience, and cost.

MULTI-CLOUD

Using services from two or more cloud providers—for example, one provider for analytics and another for backup or AI.

HYBRID CLOUD

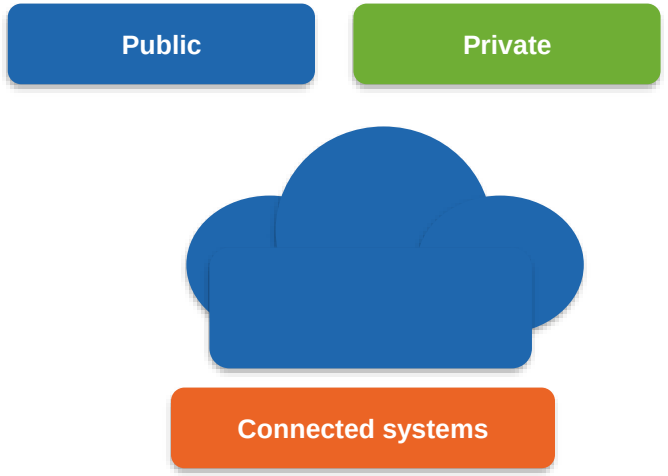
Combining private infrastructure or a private cloud with public-cloud services, linked so data and applications can work across both.

Public cloud: shared provider infrastructure delivered over the internet.

Private cloud: dedicated cloud environment for one organization.

Hybrid cloud: connected private and public environments.

Multi-cloud: more than one cloud provider; it can also be part of a hybrid design.



Advantages & Business Benefits

Why enterprises adopt multi-cloud and hybrid cloud strategies

RESILIENCE

Reduce dependence on one provider and design backup or recovery options.

FLEXIBILITY

Choose services that best fit each workload, region, or performance need.

CONTROL

Keep sensitive workloads in private environments while using public-cloud scale.

COST CHOICES

Compare pricing and match capacity to demand; monitor data-transfer charges.

SCALABILITY

Expand computing and storage during busy periods without buying all hardware upfront.

INNOVATION

Access specialized databases, AI, analytics, and developer tools from different providers.

Key idea: use each environment for the workload it handles best—without losing sight of security and total cost.

Disadvantages & Challenges

More environments can mean more complexity unless they are managed consistently.

Complex management: different dashboards, policies, and service designs.

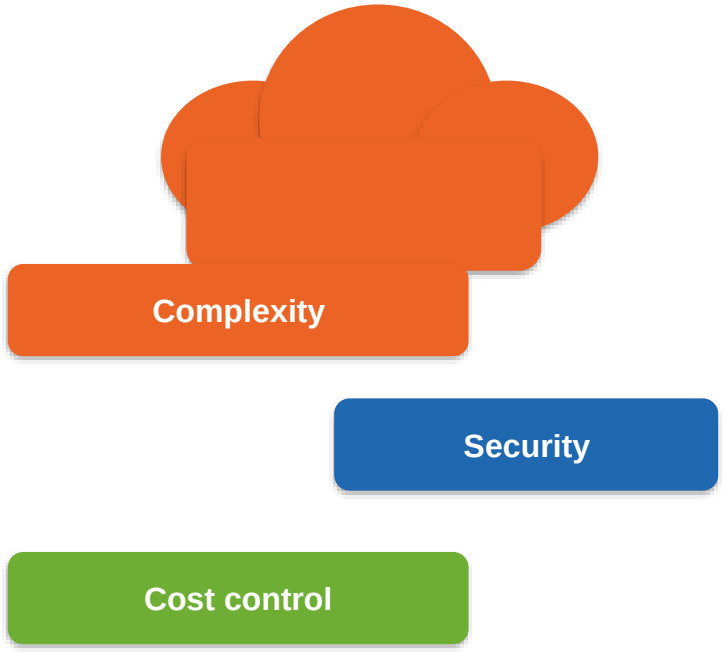
Security and identity: access rules must be consistent across every environment.

Data movement: transfer fees, latency, and integration can become costly.

Skills gap: teams need cloud, networking, automation, and security knowledge.

Vendor differences: tools and services may not work together easily.

Compliance and governance: data location, auditing, and retention must be controlled.



Mitigation: standard policies, automation, monitoring, and staff training.

Uses for Low-Income Communities

Cloud strategies can support affordable, accessible public-interest services.

LOW-COST DIGITAL LEARNING

Schools and community centers can access learning platforms without maintaining large local servers.

TELEMEDICINE & HEALTH

Clinics can securely connect records and remote consultations, where connectivity and funding allow.

DISASTER & PUBLIC SERVICES

Backups and scalable systems can keep relief information and benefit portals available during high demand.

SMALL BUSINESSES & FARMERS

Affordable online tools can help with inventory, market information, digital payments, and local selling.

Important: cloud computing does not automatically make a service free. Benefits depend on affordable internet, accessible design, privacy safeguards, local-language support, and public or community funding.

Applications & Practical Examples

Example enterprise architecture: select the right environment for each task.

E-commerce

Public cloud scales web traffic during sales.

Bank / hospital records

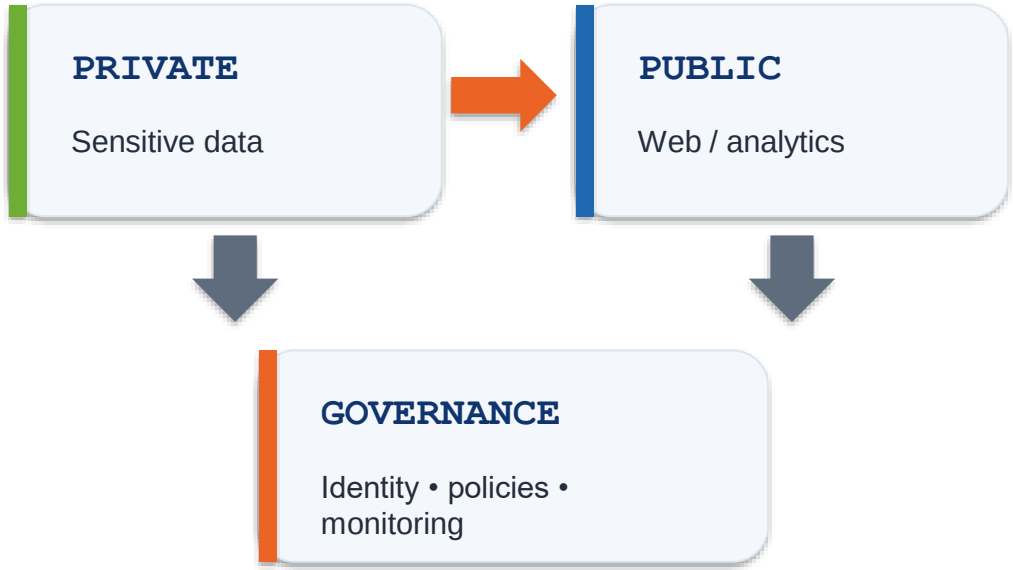
Private or tightly controlled environment for sensitive data.

Backup & disaster recovery

A second cloud or region stores recoverable copies.

AI & data analytics

Use specialized cloud services when needed.



Practical rule: classify data first, then choose the location, provid

Conclusion & Thank You

Multi-cloud and hybrid cloud are strategies—not goals by themselves.

CONCLUSION

They can improve flexibility, resilience, and access to useful services. Success depends on clear architecture, secure identity, cost monitoring, governance, and trained teams.

Start with business needs and data sensitivity.

Use common security and monitoring standards.

Measure cost, reliability, and user outcomes regularly.

THANK YOU!

Questions?

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